Ref. No.: Ex/EE/PC/B/T/21/2022

B.E. Electrical Engineering Second Year Second Semester Examination 2022
Subject: Electrical Instrumentation
Part-II: Comprehensive Semester-Exam Style Solution

Answer all questions including OR alternatives.

Prepared in a detailed, step-by-step, exam-writing style with clear definitions, derivations, and labelled diagrams.

Contents

Solution to Question 1. (a)

Problem Statement: Show that poles of a Chebyshev filter are situated on an ellipse in the s-plane.

Answer

A filter pole is a value of complex frequency s at which the denominator of the transfer function becomes zero. In the s-plane,

$$s = \sigma + j\omega,$$

where σ is the real part and ω is the imaginary-frequency part. A stable analog filter uses only poles lying in the left half of the s-plane, i.e. $\sigma < 0$.

A Chebyshev low-pass filter is designed to obtain a sharper transition from pass-band to stop-band than a Butterworth filter. Its pass-band has equal ripple. The normalized magnitude-squared response of a Chebyshev low-pass filter is

$$|H(j\omega)|^2 = \frac{1}{1 + \epsilon^2 T_n^2(\omega)},$$

where ϵ is the ripple factor and $T_n(\omega)$ is the Chebyshev polynomial of order n.

1. Chebyshev pole expression

For a normalized Chebyshev filter, the stable poles may be written as

$$s_k = -\sinh\alpha\,\sin\theta_k + j\cosh\alpha\,\cos\theta_k,$$

where

$$\theta_k = \frac{(2k-1)\pi}{2n}, \qquad k = 1, 2, \ldots, n,$$

and

$$\alpha = \frac{1}{n}\sinh^{-1}\left(\frac{1}{\epsilon}\right).$$

Let

$$s_k = \sigma_k + j\omega_k.$$

Comparing real and imaginary parts,

$$\sigma_k = -\sinh\alpha\,\sin\theta_k,$$

and

$$\omega_k = \cosh\alpha\,\cos\theta_k.$$

2. Derivation of ellipse equation

From the real part,

$$\frac{\sigma_k}{\sinh\alpha} = -\sin\theta_k.$$

Squaring,

$$\left(\frac{\sigma_k}{\sinh\alpha}\right)^2 = \sin^2\theta_k.$$

From the imaginary part,

$$\frac{\omega_k}{\cosh\alpha} = \cos\theta_k.$$

Squaring,

$$\left(\frac{\omega_k}{\cosh\alpha}\right)^2 = \cos^2\theta_k.$$

Adding the two equations,

$$\left(\frac{\sigma_k}{\sinh\alpha}\right)^2 + \left(\frac{\omega_k}{\cosh\alpha}\right)^2 = \sin^2\theta_k + \cos^2\theta_k.$$

Since

$$\sin^2\theta_k + \cos^2\theta_k = 1,$$

we get

$$\boxed{\left(\frac{\sigma}{\sinh\alpha}\right)^2 + \left(\frac{\omega}{\cosh\alpha}\right)^2 = 1}$$

This is the standard equation of an ellipse. Therefore, the Chebyshev poles lie on an ellipse in the s-plane. The stable filter chooses only the left-half-plane poles.

3. Labelled pole-location diagram

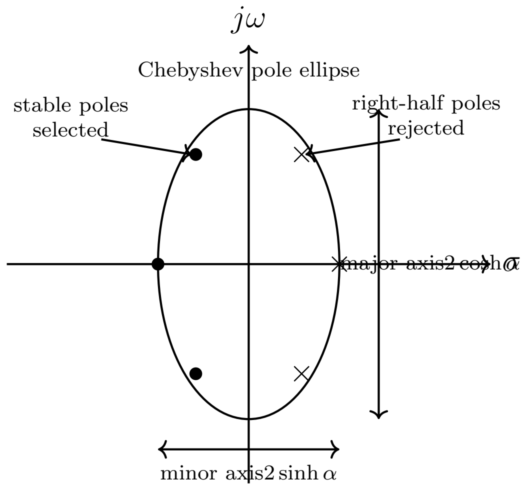

Figure 1: Chebyshev poles lie on an ellipse in the s-plane.

Final Answer

The Chebyshev pole equation reduces to

$$\boxed{\left(\frac{\sigma}{\sinh\alpha}\right)^2 + \left(\frac{\omega}{\cosh\alpha}\right)^2 = 1}$$

which is the equation of an ellipse. Hence, Chebyshev poles are situated on an ellipse in the s-plane.

Solution to Question 1. (b)

Problem Statement: Realize the filter transfer function with VCVS:

$$H(s) = \frac{60 \times 10^7}{s^2 + 3000s + 6 \times 10^7}.$$

Identify the attenuation in dB at frequencies $\omega_c/2$ and $2\omega_c$, where ω_c is the cut-off frequency.

Answer

The given transfer function is

$$H(s) = \frac{60 \times 10^7}{s^2 + 3000s + 6 \times 10^7}.$$

Since the numerator is constant and the denominator is second order, this is a **second order low-pass transfer function**. A general second order low-pass transfer function is written as

$$H(s) = \frac{K\omega_0^2}{s^2 + \frac{\omega_0}{Q}s + \omega_0^2},$$

where K is pass-band gain, ω_0 is natural angular frequency and Q is quality factor.

1. Comparing with standard form

Compare

$$\frac{60 \times 10^7}{s^2 + 3000s + 6 \times 10^7}$$

with

$$\frac{K\omega_0^2}{s^2 + \frac{\omega_0}{Q}s + \omega_0^2}.$$

Therefore,

$$\omega_0^2 = 6 \times 10^7.$$

Thus,

$$\omega_0 = \sqrt{6 \times 10^7} = 7745.97\,\text{rad/s}.$$

Also,

$$\frac{\omega_0}{Q} = 3000.$$

Therefore,

$$Q = \frac{\omega_0}{3000} = \frac{7745.97}{3000} = 2.582.$$

Now,

$$K\omega_0^2 = 60 \times 10^7 = 6 \times 10^8.$$

Since $\omega_0^2 = 6 \times 10^7$,

$$K = \frac{6 \times 10^8}{6 \times 10^7} = 10.$$

Hence,

$$\boxed{\omega_0 = 7745.97\,\text{rad/s}}, \qquad \boxed{Q = 2.582}, \qquad \boxed{K = 10}.$$

2. VCVS realization principle

For a Sallen-Key VCVS low-pass section with equal resistors and equal capacitors, the denominator may be controlled through the non-inverting gain of the VCVS amplifier. For the equal-component VCVS section,

$$\omega_0 = \frac{1}{RC},$$

and approximately,

$$Q = \frac{1}{3 - K_1},$$

where K_1 is the gain of the VCVS core used to set the quality factor. Therefore,

$$K_1 = 3 - \frac{1}{Q}.$$

Putting $Q = 2.582$,

$$K_1 = 3 - \frac{1}{2.582} = 3 - 0.387 = 2.613.$$

The required total pass-band gain is 10. Therefore an additional non-inverting gain stage may be used:

$$K_2 = \frac{10}{2.613} = 3.826.$$

Thus the VCVS core sets the denominator and Q, while the additional gain stage sets the final gain.

3. Choice of practical components

Choose

$$C = 0.01\,\mu\text{F} = 10^{-8}\,\text{F}.$$

Since

$$R = \frac{1}{\omega_0 C},$$

we get

$$R = \frac{1}{7745.97 \times 10^{-8}} = 12910\,\Omega.$$

Therefore,

$$\boxed{R \approx 12.9\,\mathrm{k}\Omega}, \qquad \boxed{C = 0.01\,\mu\mathrm{F}}.$$

4. VCVS realization diagram

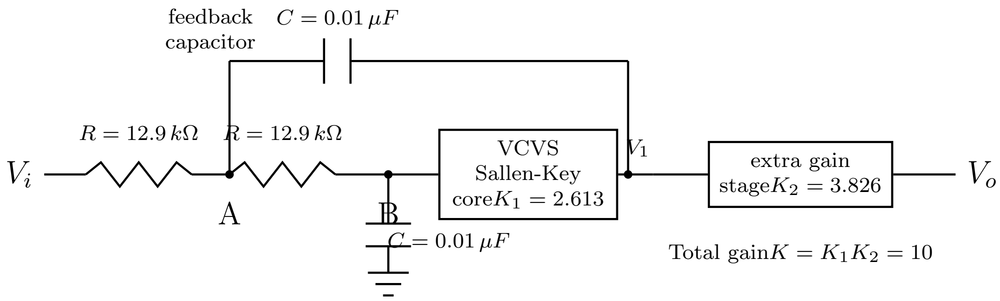

Figure 2: VCVS realization of the given second order low-pass transfer function.

5. Attenuation calculation

Let

$$\Omega = \frac{\omega}{\omega_0}.$$

For a second order low-pass filter,

$$\left|\frac{H(j\omega)}{K}\right| = \frac{1}{\sqrt{(1-\Omega^2)^2 + \left(\frac{\Omega}{Q}\right)^2}}.$$

At $\omega = \omega_c/2$

Here,

$$\Omega = \frac{1}{2} = 0.5.$$

Thus,

$$\left|\frac{H(j\omega)}{K}\right| = \frac{1}{\sqrt{(1-0.5^2)^2 + \left(\frac{0.5}{2.582}\right)^2}}.$$

$$= \frac{1}{\sqrt{(0.75)^2 + (0.1937)^2}} = \frac{1}{\sqrt{0.5625 + 0.0375}} = \frac{1}{0.7746} = 1.291.$$

The attenuation relative to pass-band gain is

$$A = 20\log_{10}\left(\frac{K}{|H|}\right) = 20\log_{10}\left(\frac{1}{1.291}\right).$$

Therefore,

$$\boxed{A(\omega_c/2) = -2.22\,\text{dB}}.$$

The negative sign indicates peaking above the nominal pass-band gain due to high Q.

At $\omega = 2\omega_c$

Here,

$$\Omega = 2.$$

Thus,

$$\left|\frac{H(j\omega)}{K}\right| = \frac{1}{\sqrt{(1-2^2)^2 + \left(\frac{2}{2.582}\right)^2}}.$$

$$= \frac{1}{\sqrt{(-3)^2 + (0.7746)^2}} = \frac{1}{\sqrt{9+0.600}} = \frac{1}{3.098} = 0.3228.$$

Hence attenuation is

$$A = 20\log_{10}\left(\frac{1}{0.3228}\right) = 9.82\,\text{dB}.$$

Therefore,

$$\boxed{A(2\omega_c) = 9.82\,\text{dB}}.$$

Final Answer

The given transfer function has

$$\boxed{\omega_0 = 7745.97\,\text{rad/s}}, \qquad \boxed{Q = 2.582}, \qquad \boxed{K = 10}.$$

A possible VCVS realization uses $R = 12.9\,\text{k}\Omega$, $C = 0.01\,\mu\text{F}$, a VCVS core gain $K_1 = 2.613$, and an extra gain stage $K_2 = 3.826$. The attenuations are

$$\boxed{A(\omega_c/2) = -2.22\,\text{dB}}, \qquad \boxed{A(2\omega_c) = 9.82\,\text{dB}}.$$

Solution to Question 2. (a)

Problem Statement: The input resistance and capacitance of a Cathode Ray Oscilloscope are $1.0\,\mu$F and $2.0\,\mathrm{M\Omega}$. What is the matched capacitance of the probe for proper compensation assuming $10\,\mathrm{M\Omega}$ probe resistance? Explain necessary formula.

Answer

A CRO does not have purely resistive input. It has an input resistance and also an input capacitance. Similarly, a probe has resistance and capacitance. Therefore the probe and CRO input form a frequency-dependent voltage divider. If it is not properly compensated, high-frequency components and low-frequency components are not attenuated equally, causing waveform distortion.

1. Equivalent circuit

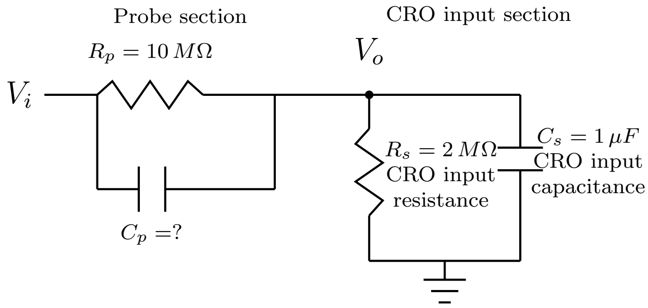

Figure 3: Equivalent circuit of compensated CRO probe and CRO input.

2. Compensation formula

For proper compensation, the time constants must be equal:

$$\boxed{R_pC_p = R_sC_s}.$$

This condition makes the attenuation independent of frequency. Therefore,

$$C_p = \frac{R_sC_s}{R_p}.$$

3. Calculation

Given,

$$R_s = 2.0\,\mathrm{M\Omega} = 2 \times 10^6\,\Omega,$$

$$C_s = 1.0\,\mu\mathrm{F} = 1 \times 10^{-6}\,\mathrm{F},$$

$$R_p = 10\,\mathrm{M\Omega} = 10 \times 10^6\,\Omega.$$

Thus,

$$C_p = \frac{(2 \times 10^6)(1 \times 10^{-6})}{10 \times 10^6}.$$

$$C_p = \frac{2}{10 \times 10^6} = 0.2 \times 10^{-6}\,\mathrm{F}.$$

Therefore,

$$\boxed{C_p = 0.2\,\mu\mathrm{F}}.$$

Final Answer

The matched capacitance of the probe is

$$\boxed{0.2\,\mu\mathrm{F}}.$$

Solution to Question 2. (b)

Problem Statement: Explain the operation of Successive Approximation Register (SAR) type Analog to Digital Converter (ADC) with a flow chart.

Answer

An ADC converts an analog voltage into a digital binary number. A Successive Approximation Register ADC is one of the most commonly used ADCs because it gives good speed with moderate circuit complexity. It works by performing a binary search on the input voltage.

The main idea is simple: first test the most significant bit, then the next bit, then the next bit, until the least significant bit is tested. At each step, the DAC output is compared with the unknown analog input.

1. Blocks of SAR ADC

1. **Sample and hold circuit:** Holds the input voltage constant during conversion.
2. **Comparator:** Compares analog input V_{in} with DAC output V_{DAC}.
3. **SAR logic/register:** Tries the bits one by one from MSB to LSB.
4. **DAC:** Converts the trial digital word into analog voltage.
5. **Clock:** Controls the bit-by-bit operation.

2. Block diagram

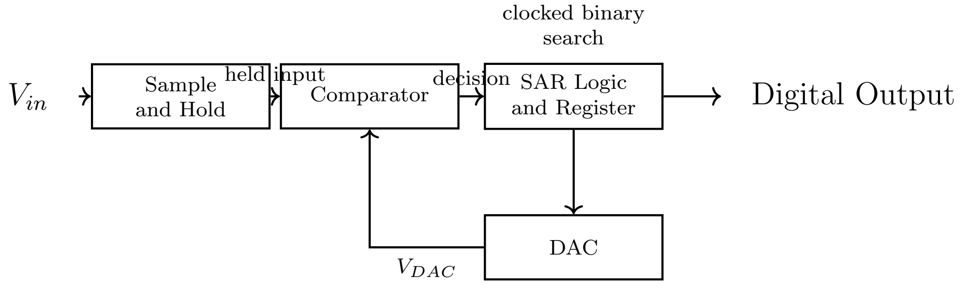

Figure 4: Block diagram of SAR ADC.

3. Flow chart of operation

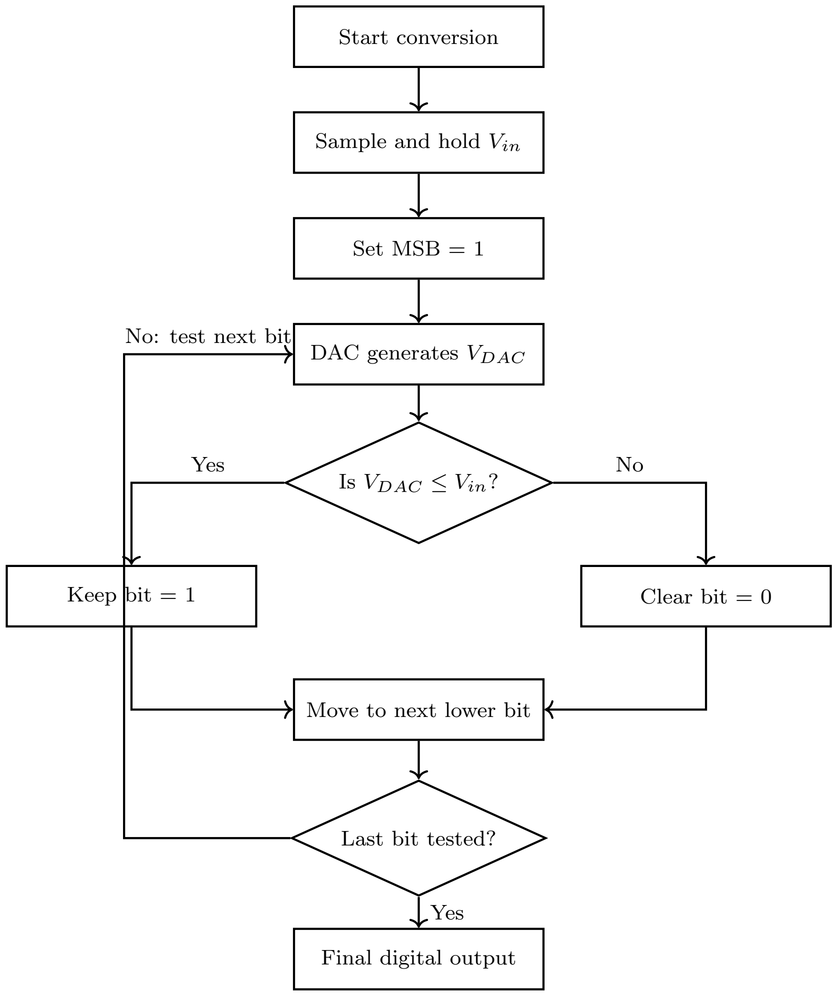

Figure 5: Flow chart of SAR ADC operation.

4. Explanation from absolute zero

Suppose an n-bit SAR ADC is used. It starts with the MSB because the MSB contributes half of full-scale value. If setting the MSB makes the DAC output greater than the input, then the MSB is too large and is cleared. If the DAC output is still less than or equal to input, the MSB is kept. The same process is repeated for each lower bit. Thus, after n clock cycles, an n-bit digital output is obtained.

For an n-bit ADC, the conversion time is approximately

$$T_c = nT_{clk},$$

where T_{clk} is the clock period.

Final Answer

A SAR ADC uses a comparator, SAR register and DAC to perform a binary search. It tests bits from MSB to LSB and gives an n-bit output in n clock periods. It is faster than counter-type ADC and widely used for instrumentation.

Solution to Question 3. (a)

Problem Statement: What are the advantages of switched capacitor circuits for implementing active electrical filters? How can you realize a band-pass filter using switched capacitor circuits?

Answer

A switched capacitor circuit replaces a resistor by a capacitor and MOS switches. If a capacitor C_s is alternately connected between two nodes at a clock frequency f_s, the average charge transfer behaves like current through a resistor. The equivalent resistance is

$$\boxed{R_{eq} = \frac{1}{f_s C_s}}.$$

1. Advantages of switched capacitor filters

1. **Suitable for IC technology:** Large resistors occupy large chip area. Capacitors and MOS switches are easier to fabricate in IC form.
2. **Accurate time constants:** In ICs, capacitor ratios are more accurate than resistor values. Therefore accurate filter response is obtained.
3. **Electronic tuning:** Since $R_{eq} = 1/(f_s C_s)$, the equivalent resistance is controlled by clock frequency.
4. **Large resistance simulation:** A small capacitor with suitable clock frequency can simulate a very large resistor.
5. **Good matching:** Capacitor matching in IC technology is better than absolute resistor matching.
6. **Useful for active filters:** Low-pass, high-pass, band-pass and band-stop filters can be realized using switched capacitor integrators.

2. Basic switched-capacitor resistor equivalent

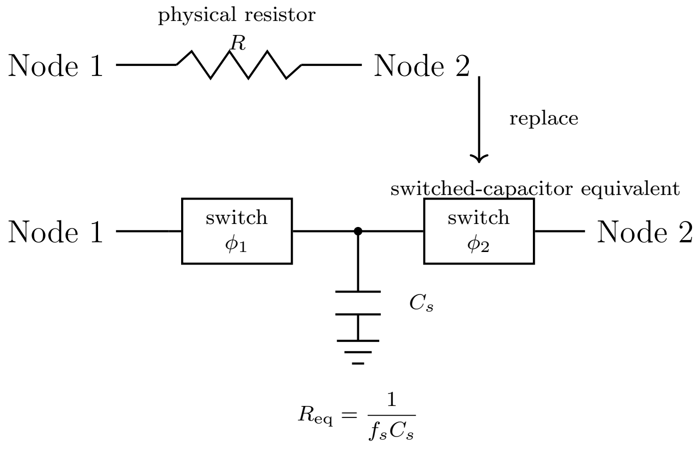

Figure 6: A resistor implemented by a switched capacitor.

3. Band-pass realization using switched-capacitor integrators

A second-order band-pass filter can be realized from a state-variable structure. The high-pass output is integrated once to produce band-pass output and integrated again to produce low-pass output. In a switched-capacitor IC realization, the resistors inside integrators and summers are replaced by switched capacitor equivalents.

For a second-order band-pass response,

$$H_{BP}(s) = \frac{K_B s}{s^2 + \frac{\omega_0}{Q}s + \omega_0^2}.$$

In state-variable form,

$$H_{BP}(s) = \frac{-K_1 K_2 s}{s^2 + K_2 K_4 s + K_2 K_3}.$$

Comparing,

$$\omega_0^2 = K_2 K_3,$$

$$\frac{\omega_0}{Q} = K_2 K_4.$$

Thus,

$$\boxed{\omega_0 = \sqrt{K_2 K_3}}, \qquad \boxed{Q = \frac{\sqrt{K_2 K_3}}{K_2 K_4}}.$$

4. Labelled switched-capacitor band-pass block diagram

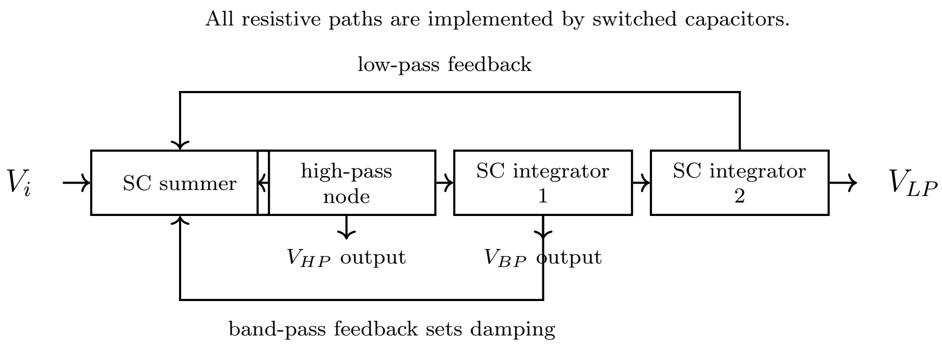

Figure 7: Band-pass filter realization using switched-capacitor state-variable structure.

Final Answer

Switched-capacitor circuits are preferred in IC active filters because they avoid large resistors, give accurate time constants, are tunable by clock frequency, and are compact. A band-pass filter is realized using switched-capacitor integrators and a summer, with the output taken after the first integrator.

Solution to Question 3. (b)

Problem Statement: Explain lock range and capture range of a Phase Locked Loop (PLL). How can you use a PLL as frequency demodulator?

Answer

A Phase Locked Loop is a closed-loop feedback system in which the output frequency and phase of a voltage controlled oscillator are made to follow an input signal. A PLL contains a phase detector, a low-pass filter and a voltage controlled oscillator.

1. Basic PLL block diagram

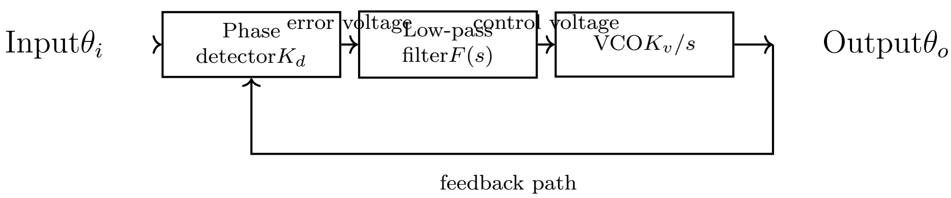

Figure 8: Basic block diagram of a PLL.

2. Lock range

Lock range is the range of input frequencies over which the PLL remains locked once it has already acquired lock. In locked condition, the VCO frequency is exactly equal to the input signal frequency, but there may be a constant phase difference.

If ω_0 is the free-running angular frequency of the VCO, then lock range is generally expressed as

$$\omega_0 - \Delta\omega_L \leq \omega_i \leq \omega_0 + \Delta\omega_L.$$

Here $\Delta\omega_L$ is the lock range on either side of the free-running frequency.

3. Capture range

Capture range is the range of input frequencies over which the PLL can acquire lock starting from an unlocked condition. The capture range is usually smaller than lock range because the loop filter must first produce a control voltage capable of pulling the VCO toward the input frequency.

$$\boxed{\text{Capture range} < \text{Lock range}}$$

4. Range diagram

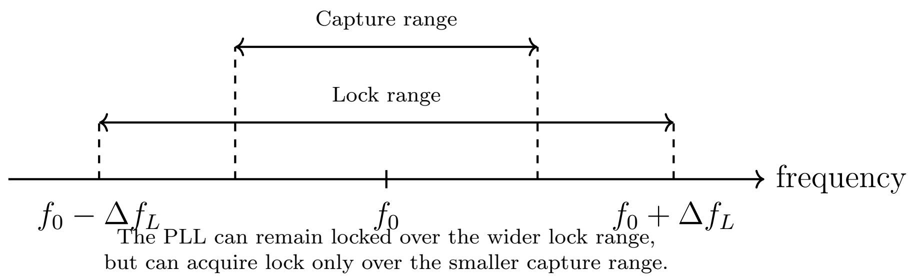

Figure 9: Lock range and capture range of a PLL.

5. PLL as frequency demodulator

In FM, the instantaneous input angular frequency is

$$\omega_i(t) = \omega_c + k_f m(t),$$

where $m(t)$ is the modulating signal. For a VCO,

$$\omega_o(t) = \omega_{FR} + K_v V_c(t).$$

For demodulation, the VCO free-running frequency is adjusted equal to carrier frequency:

$$\omega_{FR} = \omega_c.$$

In locked condition,

$$\omega_o(t) = \omega_i(t).$$

Therefore,

$$\omega_c + K_v V_c(t) = \omega_c + k_f m(t).$$

Cancelling ω_c,

$$K_v V_c(t) = k_f m(t).$$

Thus,

$$\boxed{V_c(t) = \frac{k_f}{K_v} m(t)}.$$

The control voltage is proportional to the original modulating signal. Therefore the loop filter output is the demodulated output.

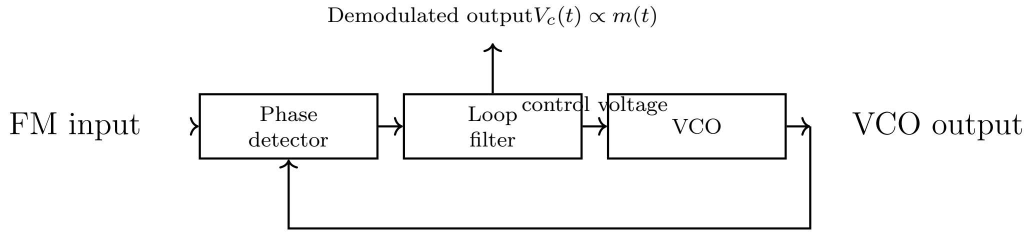

Figure 10: PLL used as a frequency demodulator.

Final Answer

Lock range is the frequency range over which a PLL remains locked. Capture range is the frequency range over which it can acquire lock. A PLL works as a frequency demodulator because its loop-filter output is proportional to the frequency deviation of the FM signal.

Solution to Question 4. (a)

Problem Statement: Explain with circuit diagram the operation of an R-2R ladder Digital to Analog Converter (DAC) considering 3 bits.

Answer

A DAC converts a digital binary input into an analog voltage or current. In an R-2R ladder DAC, only two resistor values, R and $2R$, are used. This makes it more practical than a binary-weighted resistor DAC, especially for higher number of bits.

For a 3-bit DAC, let the digital input be

$$b_2b_1b_0,$$

where b_2 is MSB and b_0 is LSB.

1. Circuit diagram

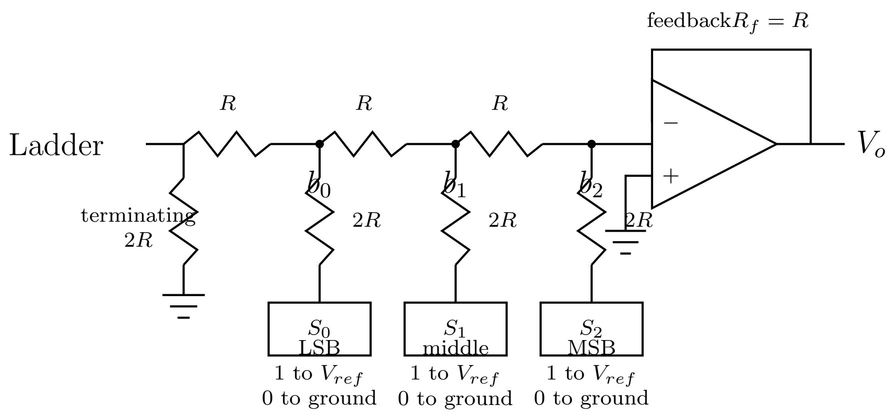

Figure 11: 3-bit R-2R ladder DAC with op-amp output.

2. Operation

Each switch connects its branch either to V_{ref} for logic 1 or to ground for logic 0. The R-2R ladder divides currents in binary ratios. The MSB contributes maximum current, the next bit contributes half of that, and the LSB contributes still smaller current.

For a 3-bit inverting R-2R DAC with $R_f = R$,

$$V_o = -V_{ref}\left(\frac{b_2}{2} + \frac{b_1}{4} + \frac{b_0}{8}\right).$$

Magnitude form:

$$\boxed{|V_o| = V_{ref}\left(\frac{b_2}{2} + \frac{b_1}{4} + \frac{b_0}{8}\right)}.$$

3. Output table

Input	Decimal	$\lvert V_o \rvert$
$hline$000	0	0
001	1	$V_{ref}/8$
010	2	$2V_{ref}/8$
011	3	$3V_{ref}/8$
100	4	$4V_{ref}/8$
101	5	$5V_{ref}/8$
110	6	$6V_{ref}/8$
111	7	$7V_{ref}/8$

Final Answer

The 3-bit R-2R ladder DAC uses only R and $2R$ resistors. Its output is the analog equivalent of binary input:

$$\boxed{V_o = -V_{ref}\left(\frac{b_2}{2} + \frac{b_1}{4} + \frac{b_0}{8}\right)}.$$

Solution to Question 4. (b)

Problem Statement: The reference voltage of a unipolar 8-bit DAC is 10 V. An offset error of −0.5 LSB exists in the DAC. If all zeroes represent 0 V without this error, what outputs are produced for input code 10110101 with and without this offset?

Answer

For an 8-bit unipolar DAC,

$$2^8 = 256.$$

The value of one LSB is

$$1\,\mathrm{LSB} = \frac{V_{ref}}{2^8} = \frac{10}{256} = 0.0390625\,\mathrm{V}.$$

1. Decimal equivalent of input code

Given digital input:

$$10110101_2.$$

Converting into decimal,

$$D = 1 \times 2^7 + 0 \times 2^6 + 1 \times 2^5 + 1 \times 2^4 + 0 \times 2^3 + 1 \times 2^2 + 0 \times 2^1 + 1 \times 2^0.$$

$$D = 128 + 0 + 32 + 16 + 0 + 4 + 0 + 1 = 181.$$

Therefore,

$$\boxed{10110101_2 = 181_{10}}.$$

2. Output without offset error

For ideal DAC,

$$V_o = D \times 1\,\mathrm{LSB}.$$

Thus,

$$V_o = 181 \times 0.0390625 = 7.0703125\,\mathrm{V}.$$

Therefore,

$$\boxed{V_o = 7.0703\,\mathrm{V} \quad \text{without offset error}}.$$

3. Output with offset error

Offset error is

$$-0.5\,\text{LSB} = -0.5 \times 0.0390625 = -0.01953125\,\text{V}.$$

Therefore,

$$V_o' = 7.0703125 - 0.01953125 = 7.05078125\,\text{V}.$$

Thus,

$$\boxed{V_o' = 7.0508\,\text{V} \quad \text{with } -0.5\,\text{LSBoffset}}.$$

Final Answer

$$\boxed{\text{Without offset error: } 7.0703\,\text{V}}$$

$$\boxed{\text{With } -0.5 \text{ LSB offset error: } 7.0508\,\text{V}}$$

Solution to Question 5. (a)

Problem Statement: Write short note on representation of errors for ADCs.

Answer

An ADC should ideally convert an analog input voltage into the nearest digital code according to a straight transfer characteristic. Practical ADCs deviate from the ideal characteristic because of offset, gain, non-linearity, comparator error, reference voltage error and quantization.

1. Quantization error

An ADC has finite resolution. Therefore a continuous input range is divided into discrete steps. The maximum quantization error is

$$\boxed{\pm\frac{1}{2}\,\text{LSB}}.$$

It is inherent even in an ideal ADC.

2. Offset error

Offset error means the entire transfer characteristic is shifted left or right from the ideal position. It is measured at the first transition point.

3. Gain error

After offset error is removed, gain error is the difference between the ideal full-scale transition and actual full-scale transition.

4. Integral non-linearity error

Integral non-linearity error is the maximum deviation of actual transition points from the ideal straight line after removing offset and gain errors.

5. Differential non-linearity error

Differential non-linearity error measures how much an actual code width differs from one ideal LSB:

$$\boxed{DNL = \text{actual code width} - 1\,\text{LSB}}.$$

If DNL is less than -1 LSB, missing codes can occur.

6. Diagram of ADC errors

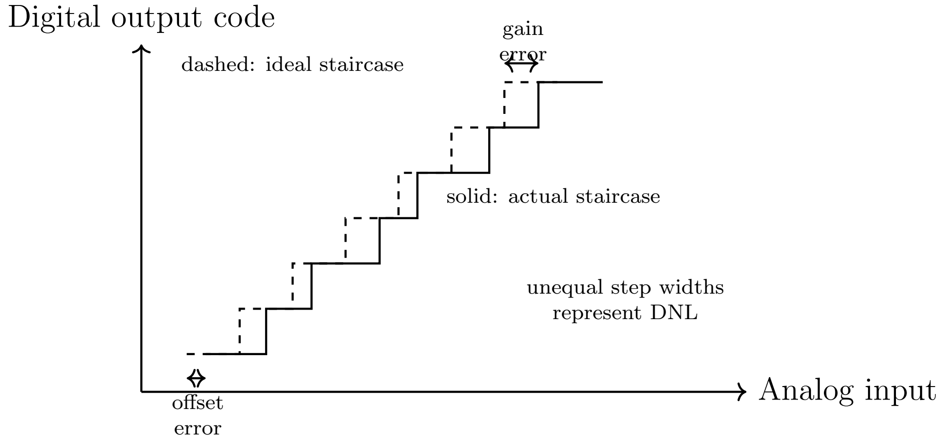

Figure 12: Representation of ADC offset, gain and non-linearity errors.

Final Answer

ADC errors are represented in terms of LSB. The important errors are quantization error, offset error, gain error, integral non-linearity, differential non-linearity and missing-code error.

Solution to Question 5. (b)

Problem Statement: Write short note on the linear model of Phase Locked Loop (PLL).

Answer

For small phase error, a PLL can be represented by a linear feedback model. This model is useful for deriving closed-loop transfer function, error transfer function and dynamic response.

1. Linear blocks

The phase detector produces voltage proportional to phase error:

$$V_e(s) = K_d[\theta_i(s) - \theta_o(s)].$$

The loop filter has transfer function $F(s)$:

$$V_c(s) = F(s)V_e(s).$$

The VCO converts control voltage into frequency. Since frequency is derivative of phase,

$$\theta_o(s) = \frac{K_v}{s}V_c(s).$$

2. Linear model diagram

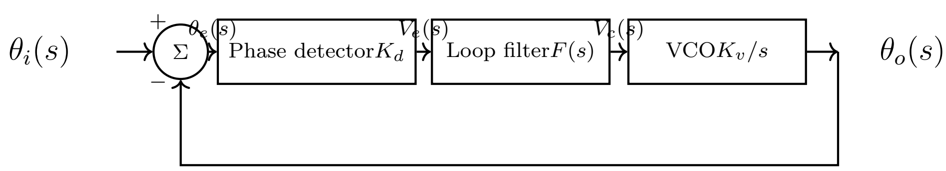

Figure 13: Linear phase-domain model of PLL.

3. Closed loop transfer function

From the block diagram,

$$\theta_o(s) = \frac{K_v}{s}F(s)K_d[\theta_i(s) - \theta_o(s)].$$

Let

$$K = K_dK_v.$$

Then

$$\theta_o(s) = \frac{KF(s)}{s}[\theta_i(s) - \theta_o(s)].$$

Multiplying by s,

$$s\theta_o(s) = KF(s)\theta_i(s) - KF(s)\theta_o(s).$$

$$\theta_o(s)[s + KF(s)] = KF(s)\theta_i(s).$$

Therefore,

$$\boxed{\frac{\theta_o(s)}{\theta_i(s)} = \frac{KF(s)}{s + KF(s)}}.$$

4. Error transfer function

The phase error is

$$\theta_e(s) = \theta_i(s) - \theta_o(s).$$

Thus,

$$\frac{\theta_e(s)}{\theta_i(s)} = 1 - \frac{\theta_o(s)}{\theta_i(s)}.$$

$$\frac{\theta_e(s)}{\theta_i(s)} = 1 - \frac{KF(s)}{s + KF(s)}.$$

Therefore,

$$\boxed{\frac{\theta_e(s)}{\theta_i(s)} = \frac{s}{s + KF(s)}}.$$

Final Answer

The linear PLL model gives

$$\boxed{\frac{\theta_o(s)}{\theta_i(s)} = \frac{KF(s)}{s + KF(s)}}$$

and

$$\boxed{\frac{\theta_e(s)}{\theta_i(s)} = \frac{s}{s + KF(s)}}.$$

Solution to Question 5. (c)

Problem Statement: Write short note on active filter realizations using state-variable representation.

Answer

State-variable representation is a powerful method for realizing active filters. It uses integrators and a summing amplifier. The outputs of the integrators are treated as state variables. A very important advantage is that low-pass, high-pass and band-pass outputs can be obtained simultaneously from one circuit.

1. Standard second order denominator

A general second order denominator is

$$D(s) = s^2 + \frac{\omega_0}{Q}s + \omega_0^2.$$

The three basic second order filter responses are:

$$H_{LP}(s) = \frac{K\omega_0^2}{s^2 + \frac{\omega_0}{Q}s + \omega_0^2},$$

$$H_{BP}(s) = \frac{K_B s}{s^2 + \frac{\omega_0}{Q}s + \omega_0^2},$$

$$H_{HP}(s) = \frac{Ks^2}{s^2 + \frac{\omega_0}{Q}s + \omega_0^2}.$$

All have the same denominator. Thus, one active structure can produce all three responses.

2. State variable form

For the state-variable structure, the transfer functions can be written as

$$H_{HP}(s) = \frac{K_1 s^2}{s^2 + K_2K_4 s + K_2K_3},$$

$$H_{BP}(s) = \frac{-K_1K_2 s}{s^2 + K_2K_4 s + K_2K_3},$$

$$H_{LP}(s) = \frac{K_1K_2K_3}{s^2 + K_2K_4 s + K_2K_3}.$$

Thus,

$$\omega_0^2 = K_2K_3,$$

and

$$\frac{\omega_0}{Q} = K_2K_4.$$

Therefore,

$$\boxed{\omega_0 = \sqrt{K_2K_3}}, \qquad \boxed{Q = \frac{\sqrt{K_2K_3}}{K_2K_4}}.$$

3. State-variable block diagram

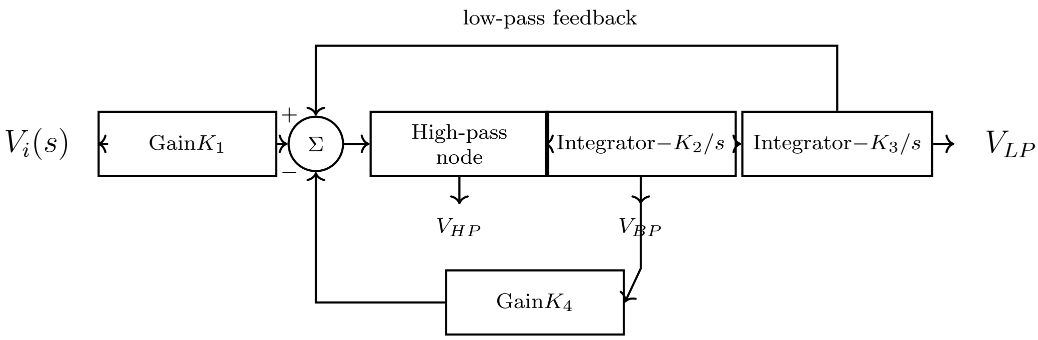

Figure 14: Active filter realization using state-variable representation.

4. Why it is called universal active filter

The same circuit gives:

$$V_{HP}, \qquad V_{BP}, \qquad V_{LP}.$$

By adding a summer to combine HP and LP outputs, a band-stop output can also be produced:

$$V_{BS} = V_{HP} + V_{LP}.$$

Therefore, a state-variable filter is also called a universal active filter.

5. Advantages

1. It gives HP, BP and LP outputs simultaneously.
2. ω_0 and Q can be adjusted independently.
3. It is suitable for second order filter sections.
4. It is useful in integrated active filter design.
5. It can be converted into switched-capacitor form.

Final Answer

Active filters using state-variable representation are realized using one summer and two integrators. The outputs are taken at different points to obtain high-pass, band-pass and low-pass responses. Since one circuit can realize several filter types, it is called a universal active filter.

Additional Exam-Oriented Derivation Bank

This section is included to make the solution more useful for semester writing. In an examination, a complete answer should not only give the final formula, but should also show why the formula appears, what each symbol means, and how the circuit or block diagram is interpreted.

A. How to identify a filter from a transfer function

A second order filter has the denominator

$$D(s) = s^2 + a_1 s + a_0.$$

The numerator decides the type of response:

Numerator form	Filter type
constant	low-pass
s	band-pass
s^2	high-pass
$s^2 + \omega_0^2$	band-stop or notch

For the VCVS problem, the numerator is constant. Therefore the circuit is a second order low-pass filter. This is the first observation that should be written in the answer script.

B. Standard comparison method

Whenever a second order transfer function is given, compare it with

$$H(s) = \frac{K\omega_0^2}{s^2 + \frac{\omega_0}{Q}s + \omega_0^2}.$$

Then directly read:

$$\omega_0^2 = \text{constant term of denominator},$$

$$\frac{\omega_0}{Q} = \text{coefficient of } s,$$

$$K = \frac{\text{constant numerator}}{\omega_0^2}.$$

This comparison is very important because it avoids blind memorisation.

C. Why SAR ADC is a binary search

In an n-bit ADC, the MSB represents half-scale. If the MSB is tested first, the ADC immediately decides whether the unknown input lies in the upper half or lower half of the range. The next bit again divides the remaining range into two parts. Therefore SAR operation is not random trial; it is a systematic binary search.

For example, for a 4-bit ADC, the trial sequence is:

$$1000 \rightarrow 1100 \text{ or } 0100 \rightarrow \cdots$$

At every clock pulse, one bit is decided permanently. Hence an n-bit SAR ADC requires nearly n clock pulses.

D. Physical meaning of probe compensation

A CRO probe compensation problem should be explained from the idea of equal time constants. If the probe time constant and CRO input time constant are not equal, the divider ratio changes with frequency. A square wave contains many high-frequency harmonics. Therefore, a mismatched probe changes the edge shape of a square wave.

The matched condition is

$$R_pC_p = R_sC_s.$$

This means:

- the resistive divider ratio remains fixed at low frequency,
- the capacitive divider ratio remains consistent at high frequency,
- the overall attenuation becomes frequency independent.

E. Physical meaning of lock range and capture range

A PLL has two different ranges because locking and acquiring lock are not the same action.

Lock range: If PLL is already locked, it can follow slow changes of input frequency over a wide range. This is called tracking.

Capture range: If PLL is initially unlocked, it must first generate a slowly varying control voltage through the loop filter. This process is harder. Therefore capture range is smaller.

Hence, in words:

$$\boxed{\text{Capture range is the acquisition range, while lock range is the tracking range.}}$$

F. Why R-2R ladder is preferred

In a binary-weighted resistor DAC, resistor values become $R, 2R, 4R, 8R, \ldots$. For high bit number, very large and very small resistor values are required, and accurate fabrication becomes difficult. In R-2R ladder DAC, only two resistor values are used. Therefore matching is easier and IC fabrication is practical.

G. Common mistake checklist

1. In Chebyshev pole problems, do not write circle. Butterworth poles lie on a circle; Chebyshev poles lie on an ellipse.
2. In SAR ADC, do not say it counts upward like counter ADC. It uses successive approximation, i.e. MSB-to-LSB trial.
3. In PLL demodulation, the demodulated signal is taken from the loop filter output, not from the VCO RF output.
4. In R-2R DAC, remember that only two resistor values are used: R and $2R$.
5. In DAC offset error, add the offset voltage to every ideal output code.
6. In ADC error representation, quantization error exists even in an ideal ADC.
7. In state-variable filter, band-pass output is taken after the first integrator, and low-pass output after the second integrator.

H. Mark-wise presentation strategy

For a 6-mark answer:

Step 1: Define the topic in two or three lines.

Step 2: Draw a labelled diagram.

Step 3: Write the governing equation.

Step 4: Derive the required formula step by step.

Step 5: Box the final answer.

For a 10-mark answer:

Step 1: Start with a clear definition.

Step 2: Draw a neat labelled block/circuit diagram.

Step 3: Explain each block or circuit part.

Step 4: Do full derivation or numerical calculation.

Step 5: Add a final interpretation in words.

I. Final formula map for quick last-hour revision

Topic	Formula
Chebyshev ellipse	$\left(\frac{\sigma}{\sinh\alpha}\right)^2 + \left(\frac{\omega}{\cosh\alpha}\right)^2 = 1$
VCVS second-order LPF	$H(s) = \frac{K\omega_0^2}{s^2+(\omega_0/Q)s+\omega_0^2}$
Probe compensation	$R_pC_p = R_sC_s$
SAR conversion time	$T_c \approx nT_{clk}$
Switched capacitor resistor	$R_{eq} = \frac{1}{f_sC_s}$
PLL demodulated output	$V_c(t) = \frac{k_f}{K_v}m(t)$
3-bit R-2R DAC	$V_o = -V_{ref}\left(\frac{b_2}{2} + \frac{b_1}{4} + \frac{b_0}{8}\right)$
ADC quantization error	$\pm\frac{1}{2}$ LSB
PLL transfer function	$\frac{\theta_o(s)}{\theta_i(s)} = \frac{KF(s)}{s+KF(s)}$
State-variable frequency	$\omega_0 = \sqrt{K_2K_3}$

Concluding Revision Sheet

Topic	Most important formula / idea
Chebyshev pole locus	$(\sigma/\sinh\alpha)^2 + (\omega/\cosh\alpha)^2 = 1$
VCVS low-pass	$H(s) = K\omega_0^2/(s^2 + (\omega_0/Q)s + \omega_0^2)$
Probe compensation	$R_pC_p = R_sC_s$
SAR ADC	Tests bits from MSB to LSB using comparator and DAC.
Switched capacitor	$R_{eq} = 1/(f_sC_s)$
PLL demodulator	$V_c(t) = (k_f/K_v)m(t)$
R-2R DAC	$V_o = -V_{ref}(b_2/2 + b_1/4 + b_0/8)$ for 3 bit.
DAC offset error	Add offset voltage to ideal DAC output.
ADC errors	Quantization, offset, gain, INL, DNL, missing code.
PLL linear model	$\theta_o/\theta_i = KF(s)/(s + KF(s))$
State-variable filter	$\omega_0 = \sqrt{K_2K_3}, \quad Q = \sqrt{K_2K_3}/(K_2K_4)$